SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2023 – Oct 2023			
Maximum Marks: 30	**Examination: In-Semester Examination**	**Duration : 1 hour & 15 mins**	
Programme code: 1 **Programme: B.Tech Computer Engineering**		**Class:** LY	**Semester:** VII **(SVU 2020)**
Name of the Constituent College: **K. J. Somaiya College of Engineering**		**Name of the department:** COMP - Honours (DSA)	
Course Code: 116h54C701	**Name of the Course: Advanced Machine Learning**		

Question No.		Max. Marks	CO Mapped	BT Level
Q1	Consider a simple linear regression problem with a single feature (x) and a target (y). You start with initial values of the coefficients m (slope) and c (intercept) as 0. The equation for the model is: $y = mx + c$. You are given the following dataset: **x** **y** 1 3 2 4 3 5 4 6 The learning rate (alpha) is set to 0.1 and the momentum parameter (beta) is set to 0.9 in the case of Momentum-based Gradient Descent. • Perform one epoch of Batch Gradient Descent. Show the updates to the coefficients after the full pass through the dataset. • Perform one epoch of Momentum-based Gradient Descent. Show the updates to the coefficients after the full pass through the dataset. **OR** Compare the representational power of deep neural networks (DNNs) with shallow networks. Discuss how the depth of a neural network impacts its ability to capture hierarchical and complex patterns in data. Provide examples of tasks where depth is crucial for superior performance.	10	CO1 & CO2	Understand & Apply

Q2	For a CNN architecture for image classification with the following specifications: • Three convolutional layers with 3x3 filters and 32 filters in each layer. • ReLU activation functions and batch normalization after each convolutional layer. • Max-pooling layers (2x2) after the first and second convolutional layers. • Fully connected layer with 128 neurons and ReLU activation. • The output layer have softmax activation for multiclass classification. a) Draw the architecture and calculate the total number of parameters in the network. b) Explain the significance of max-pooling and batch normalization in this architecture and how they contribute to model performance	10	CO3	Apply & Analysis
Q3	**Answer any TWO** a) Explain the basic architecture of a Recurrent Neural Network (RNN). Include a description of the hidden state, input sequence, and the recurrent connections. Illustrate how an RNN processes sequences and handles information over time. b) Explain Softmax activation function with example. c) Assume you are training several machine learning models on the same dataset for a regression problem. For each model, you calculate the Mean Squared Error (MSE) on the training set and on a validation set. The results are as follows: **Model** **MSE (Training)** **MSE (Validation)** A 2 20 B 10 15 C 5 7 Answer the following Questions: i. Which model would you say is likely under-fitting the data? Explain your reasoning. ii. Which model is likely overfitting the data? Explain your reasoning. iii. If you had to choose one of these models to deploy, which one would you choose and why?	10 (05 + 05)	CO1, CO2 & CO3	Understand & Apply